Lecture-27

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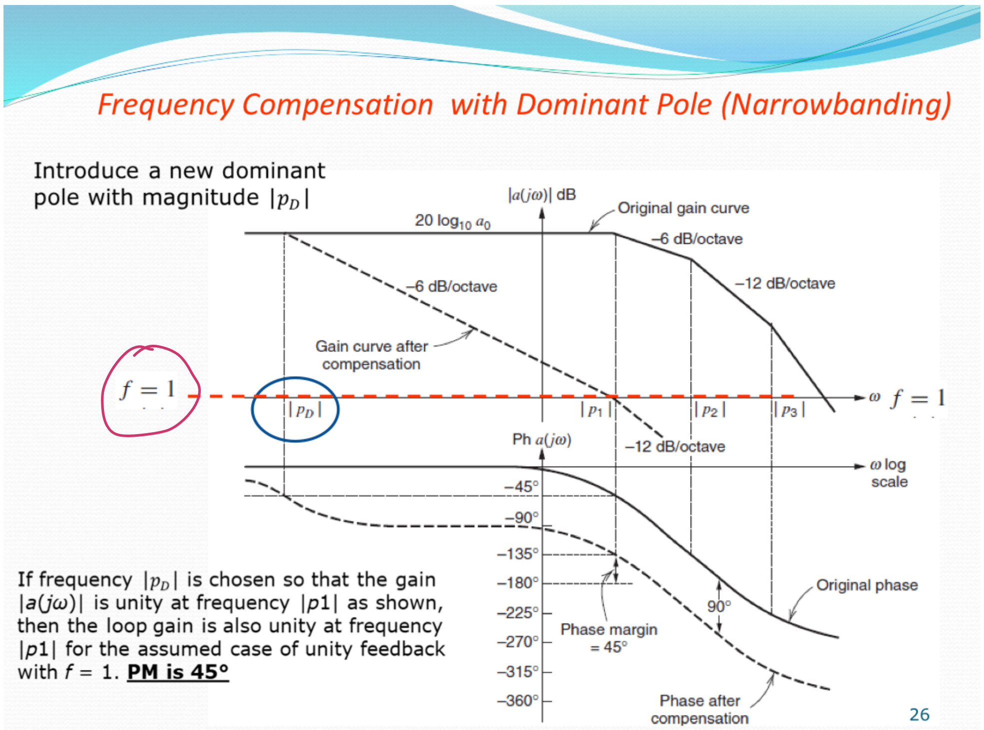
→ This is the continuation of Frequency compensation by adding Dominant pole.



→ The dominant pole change original Gain curve into a dashed line.

and a new phase margine = 45°

and the system is stable

→ We have multiple methods to achieve this goal. Another important strategy is detailed below.

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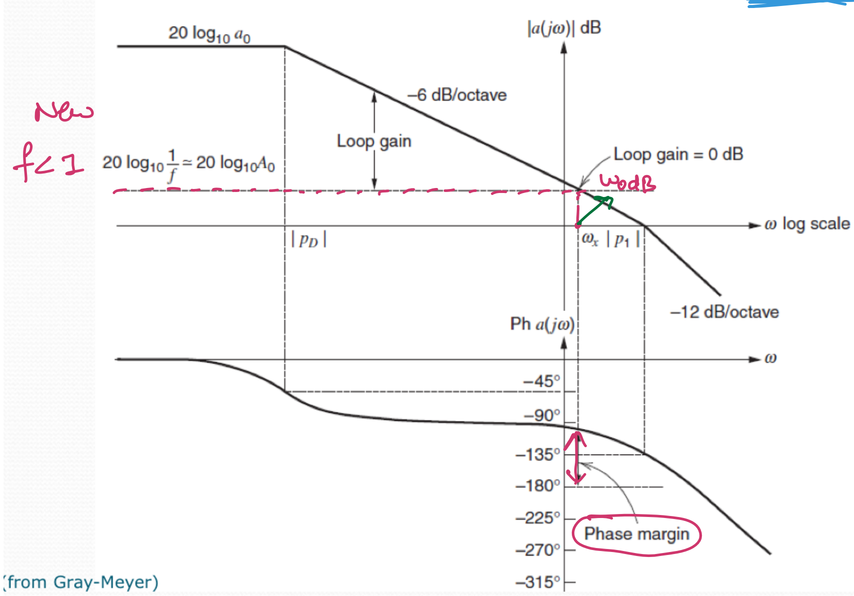
The same opamp used with a lower fedback (f<1)



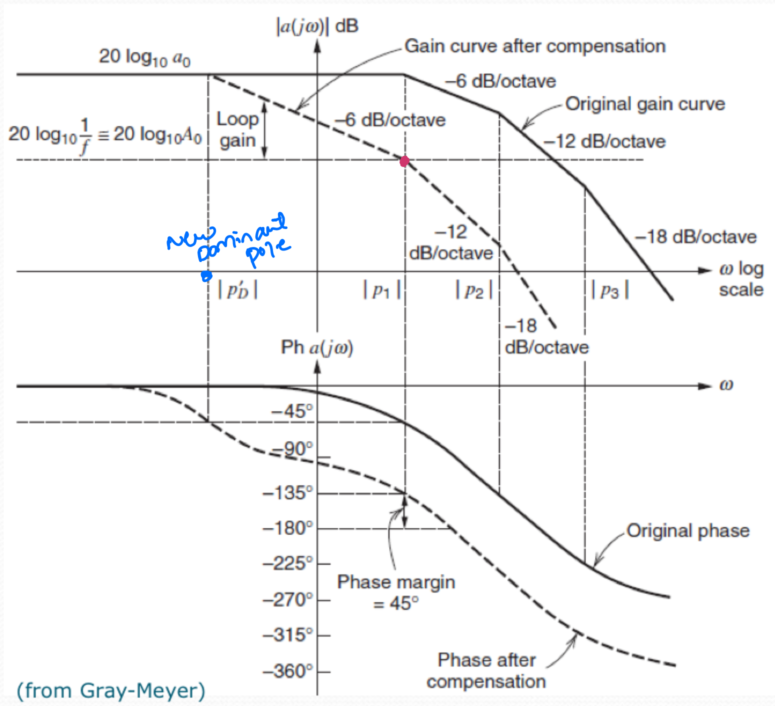
(from Gray-Meyer)

Rather than adding an extra Dominant pole, we can play with the feedback f by using $f<1$ rather than $f=1$ in the previous case.

Note:- $f=1$ is the worst possible value for 'f' in case of stability.

Because $f=1$ puts ω_{0dB} at higher angular frequencies.

The same opamp used with a lower feedback (f<1)



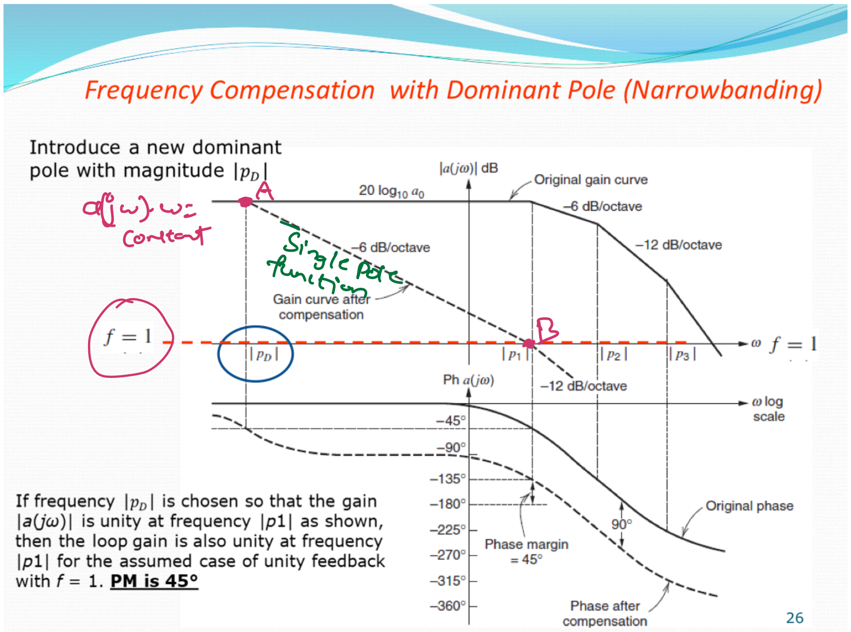
(from Gray-Meyer)

Optimum bandwidth is achieved in such circuits if the compensation is tailored to the gain value required

A dominant pole is added with magnitude $|pD'|$ to give a phase margin of 45°.
$|pD'|$ is << $|pD|$.

→ In this case even though $f<1$ and the 0dB gain of $A(j\omega)$ i.e. the dashed line $= 20\log_{10}\frac{1}{f}$
$= 20\log_{10} A_0$
is moved upwards & if we still want
$PM = 45°$?

By adding a Dominant pole as shown in the figure.



$$x = 20\log_{10}\left|a(j\omega) - 20\log_{10}\left(\frac{1}{f}\right)\right| = \text{loop gain in dB}$$

$$\boxed{x \rightarrow 20\log_{10}\left|\left(a(j\omega)\cdot f\right)\right|}$$

$$\boxed{x = 20\log_{10}\left|T(j\omega)\right|}$$

→ When we introduce a New Dominant pole P_D we take a position that upto P_1, "the gain times the frequency is constant.

i.e $\boxed{a(j\omega)\cdot\omega = \text{constant}}$

Gain · frequency

This is the case for single pole function.

$$\boxed{\text{The Gain at } A = a_0 \cdot P_D}$$

$$\boxed{\text{The Gain at B } (P_1) = a(j\omega_{P_1}) \cdot P_1}$$

$$\Downarrow$$

$$\boxed{a(j\omega_{0dB}) \cdot P_1}$$

$$\|$$

$$\boxed{\frac{1}{f} \cdot P_1} \quad (\because f = 1)$$

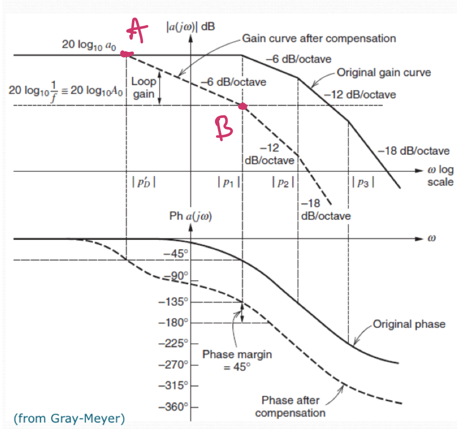
$$\Rightarrow \boxed{1 \cdot P_1}$$

$$\Rightarrow \quad a_0 P_D = P_1$$

$$\Rightarrow \quad \boxed{P_D = \frac{P}{a_0}}$$

** Similarly following this philosophy

The same opamp used with a lower feedback (f<1)



(from Gray-Meyer)

Optimum bandwidth is achieved in such circuits if the compensation is tailored to the gain value required

A dominant pole is added with magnitude $|pD'|$ to give a phase margin of 45°.
$|pD'|$ is << $|pD|$.

$$\boxed{\underbrace{\text{Gain at } A}_{\text{Gain}} \cdot \underbrace{|p'_D|}_{\text{frequency}} = \text{Gain at } B \cdot p_1}$$

Gain Bandwidth product

$$\Rightarrow \boxed{a_0 \, p'_D = \frac{1}{f} p_1}$$

$$\Rightarrow \boxed{p'_D = \frac{1 \, p_1}{f \, a_0}}$$

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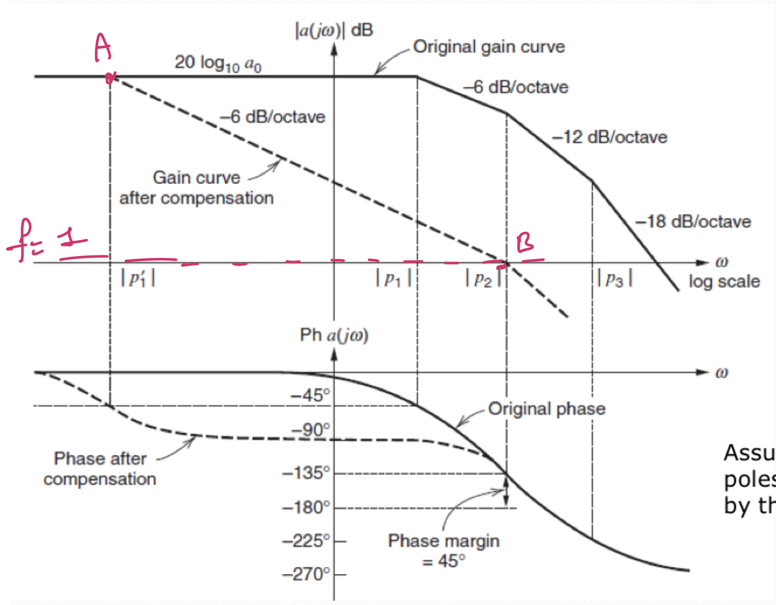
Frequency Compensation by reducing P1

(this technique requires access to the internal nodes of the amplifier)



$p'_1 = ?$

Assume that higher frequency poles $p2$ and $p3$ are unaffected by this procedure

(from Gray-Meyer)

→ Rather than adding extra pole we move the 1st pole to lower frequencies.

i.e $|P_1'|$ & $|P_2|$ are now at a distance greater than before. We have now enlarged Global Bandwidth of the circuit.

Q What should be the position of P_1' ? for phase Margin 45°.

$$a_0 P_1' = \frac{1}{f} P_2 \qquad \because f = 1$$

$$\Rightarrow \boxed{P_1' = \frac{P_2}{a_0}}$$

Reducing P_1 is a very good solution because we are not adding an extra pole.

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→ Now let's discuss it's practically achieve
"pole Compensation"

There are multiple ways.

→ If we know that a specific pole is associated with a given Node. you can add an extra Capacitor to that Node.

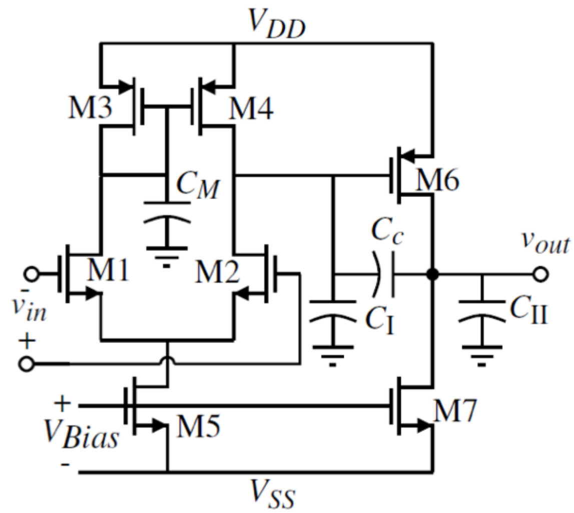
But adding extra capacitor is not very efficient because usually this pole Relocation occurs at very low frequency. To achive this we must add large Capacitance. which is quite unpractical.

→ One very effective Solution is
"Miller Compensation"

If you remember that Miller theorem states that if you have a capacitance parallel to a Circuit which realizes some kind of Gain The effect is to project capacitance at i/p Node with Multiplication factor somehow equal to the Gain.

Miller Compensation – Two stage Opamp

(P. Allen)



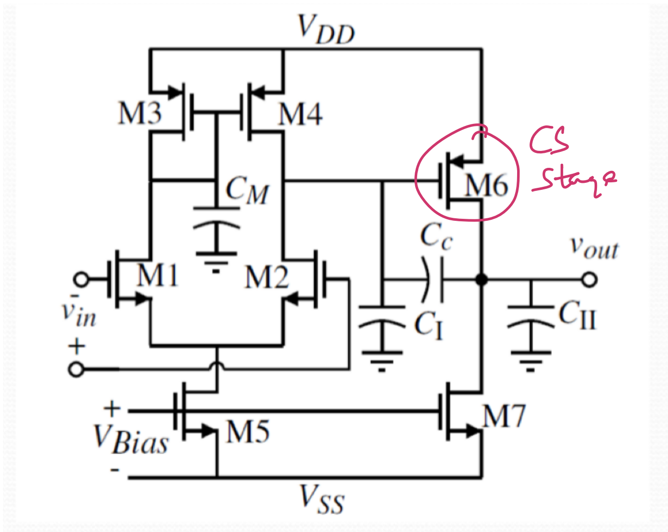
Cc = accomplishes the Miller compensation
CM = capacitance associated with the first-stage mirror (mirror pole, can be neglected)
CI = output capacitance to ground of the first-stage
CII = output capacitance to ground of the second-stage

→ ∴ Rather than placing large capacitance b/w given node to the GND, you put this capacitance parallel to gain stage. We can get large capacitances with relatively less capacitive components.

→ We will study Miller Compensation on a 2-stage Differential Amplifier.

→ 2 Stage op-Amp is an Amplifier in which in the "1st stage" we have a "Differential

Amplifier" & in the "2nd stage" we have a "common source amplifier".



→ Here M_1, M_2, M_3, M_4, M_5 for Diff-Amp with current mirror loading

& M6 realizes CS stage Topology

$$R_{out} = \frac{1}{g_{o6} + g_{o7}}$$

→ Frequency response of 2-stage Op-Amp

all the parasitic capacitances are collapsed into C_M, C_C, C_I, C_{II}

$$C_{II} = C_p + C_L$$

C_p → parasitic at load

C_L → load capacitance

& if the pole associated with the o/p

Node is the dominant pole. we can attach ~~an~~ addition capacitance in parallel with C_c

i.e $\quad C_c = C_{gd} + \underbrace{[C]}_{\text{extra capacitor}}$

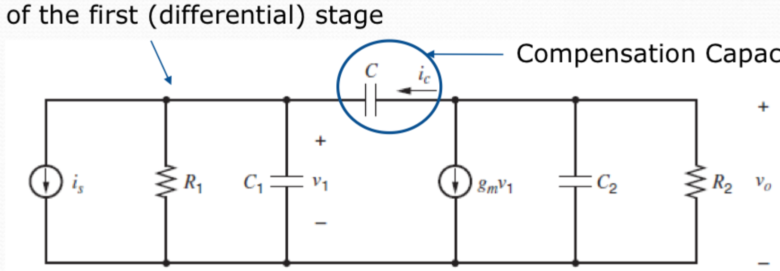
extra capacitor
miller capacitor

→ If zoom the 2nd stage and assume 1st stage as a source with simple Parasitic Resistance then we have.

Pole Splitting

Output of the first (differential) stage

Compensation Capacitance



Output of the 2nd (CS) stage

$$-i_s = \frac{v_1}{R_1} + v_1 C_1 s + (v_1 - v_o)Cs$$

$$g_m v_1 + \frac{v_o}{R_2} + v_o C_2 s + (v_o - v_1)Cs = 0$$

$$\boxed{z = \frac{g_m}{C}}$$ zero is +ve

$$\frac{v_o}{i_s} = \frac{(g_m - Cs)R_2 R_1}{1 + s[(C_2 + C)R_2 + (C_1 + C)R_1 + g_m R_2 R_1 C] + s^2 R_2 R_1 (C_2 C_1 + CC_2 + CC_1)}$$

This is similar to Frequency Response of Common Source stage.

Revice CS Frequency response

$$D(s) = \left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right)$$
$$= 1 - s\left(\frac{1}{p_1} + \frac{1}{p_2}\right) + \frac{s^2}{p_1 p_2}$$

$$p_1 = -\frac{1}{(C_2 + C)R_2 + (C_1 + C)R_1 + g_m R_2 R_1 C} \simeq -\frac{1}{g_m R_2 R_1 C}$$

gain of 2nd stage

$|p_1|$ *decreases* as C *increases*

$|p_2|$ *increases* as C *increases*

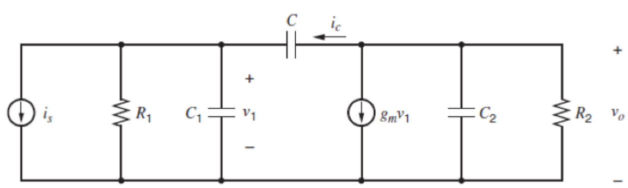
$$p_2 \simeq -\frac{g_m C}{C_2 C_1 + C(C_2 + C_1)}$$

Pole splitting

as $C \to \infty$ $\quad p_2 \to -g_m/(C_2 + C_1)$

(Gray-Meyer)

Development of the second pole



only two poles

With C shorted, C_1 and C_2 are connected in parallel $1/g_m$ is in parallel with R_2.

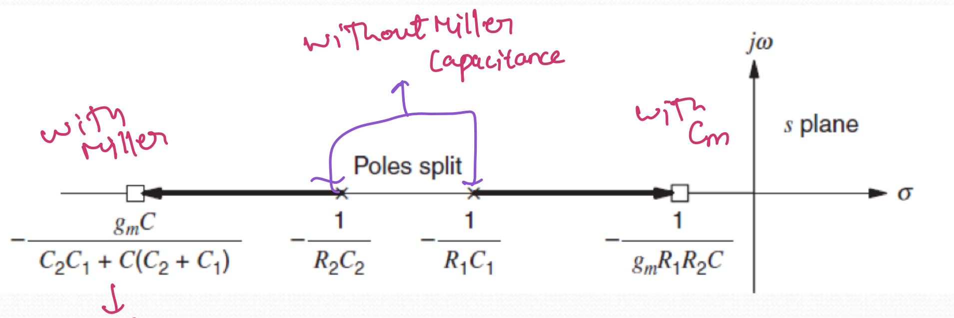
Assuming $1/g_m \ll R_2$, the short-circuit time constant

$\tau = (1/g_m)(C_1 + C_2)$ $\quad |p_2| \approx 1/\tau = g_m/(C_1 + C_2)$

for $C = 0$

$$p_1 = -\frac{1}{R_1 C_1}$$
$$p_2 = -\frac{1}{R_2 C_2}$$

(Gray-Meyer)



$\sim \frac{g_m}{C_1+C_2}$

Locus of the poles of the circuit as C is increased from zero

$$p_1 = -\frac{1}{R_1C_1}$$

$$p_2 = -\frac{1}{R_2C_2}$$

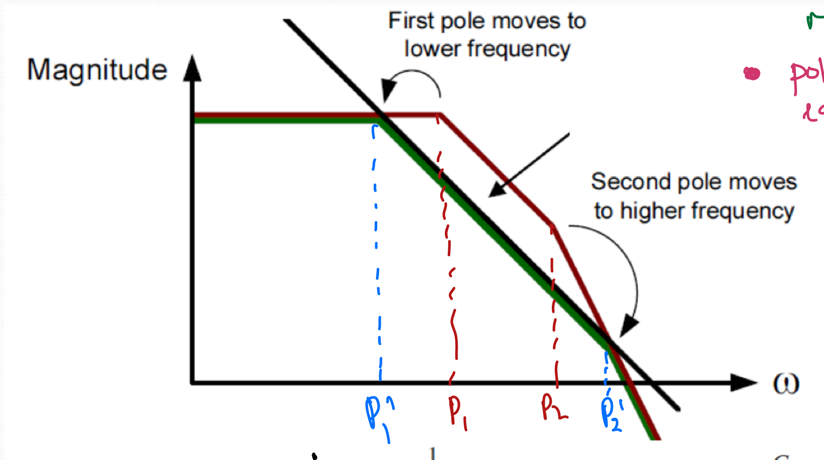
→ when ever we are analyzing stability if we have poles which are nearer to each other it is an issue.

Because Poles reduce gain by $-6\,dB/oct$ they also cause phase loss of -90° which is unwarranted.

If we have Poles well seperated the phase loss occurs quite far.

Pole Splitting

- Poles splitted because of miller
- poles close to each other



$$p_1' \simeq -\frac{1}{g_m R_2 R_1 C} \qquad p_2' \simeq -\frac{g_m C}{C_2 C_1 + C(C_2 + C_1)}$$

$$p_2 \to -g_m/(C_2 + C_1)$$

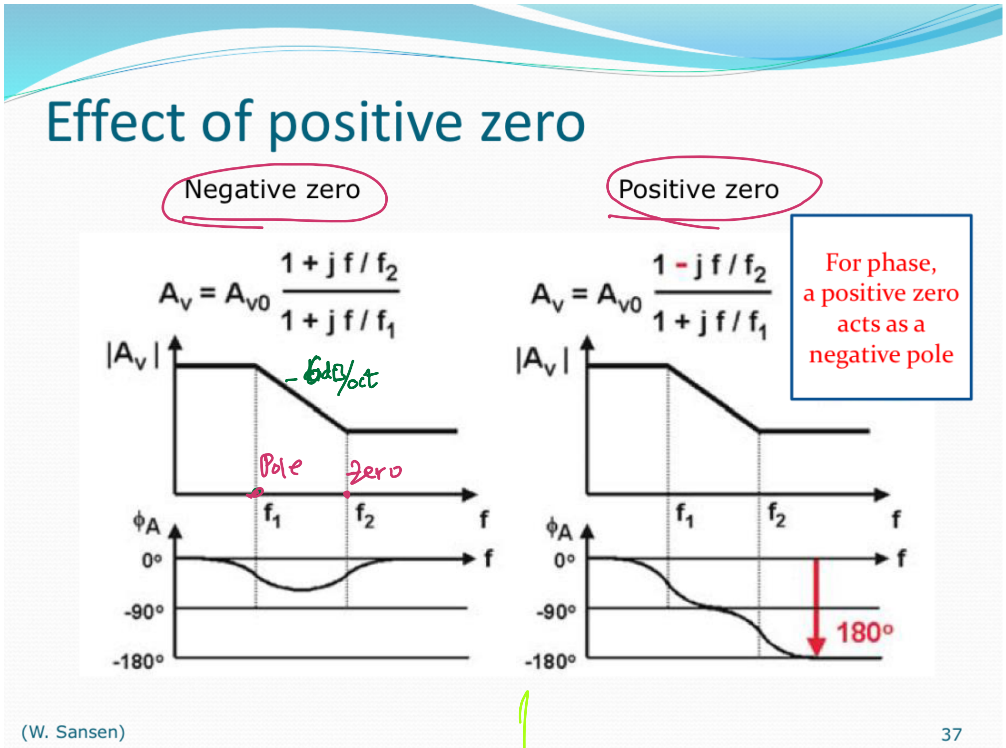
(from Gray-Meyer)

→ Pole splitting at first sight, usage of Miller effect seems to be beneficial. But there is big risk associated with the strategy because of the +ve zero we have seen in the above case.

$$\boxed{z = \frac{g_m}{C}}$$

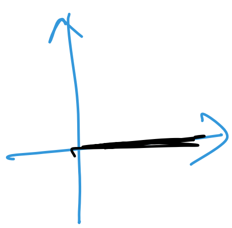
we have not considered the effect of this zero in phase margin estimation.

→ let's discuss the effects of "+ve zero" on the other hand "–ve zero" is not an issue.



At High frequencies

$$A_v = A_{v0}\left(\frac{f_1}{f_2}\right)$$

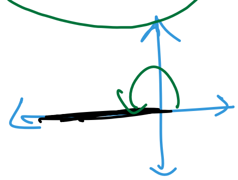


α = phase = 0°

$$A_v = A_{v0}\left(-\frac{f_1}{f_2}\right)$$

Transfer function is –ve

$\alpha = -180°$ phase



∴ +ve zero is problamatic for the phase loss

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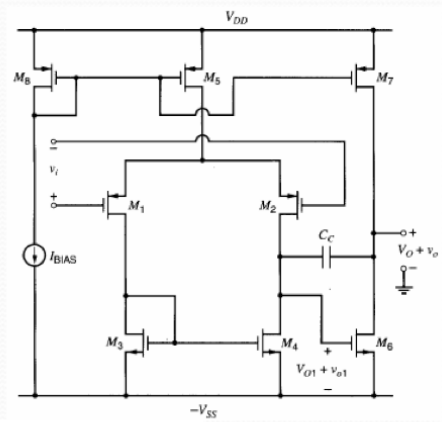
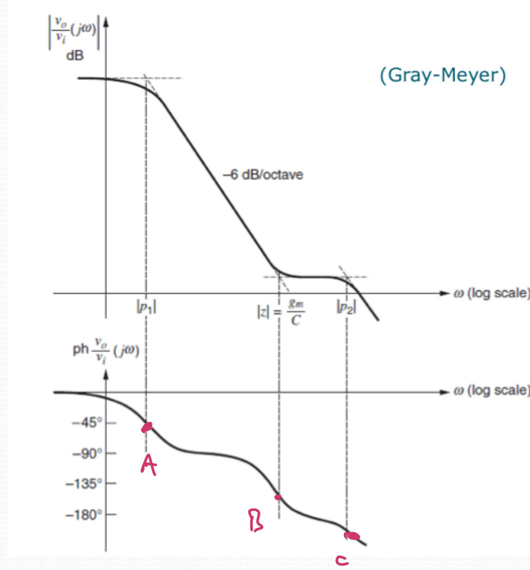
Two-Stage CMOS Opamp Compensation: the zero issue





$$p_1 \simeq -\frac{1}{g_{m_b} R_2 R_1 C_c}$$

$$\boxed{p_2 \simeq -\frac{g_m C}{C_2 C_1 + C(C_2 + C_1)}}$$

$$\boxed{z = \frac{g_{m_b}}{C}}$$

Who is bigger?

Here if we look clearly we donot have a phase Margin.

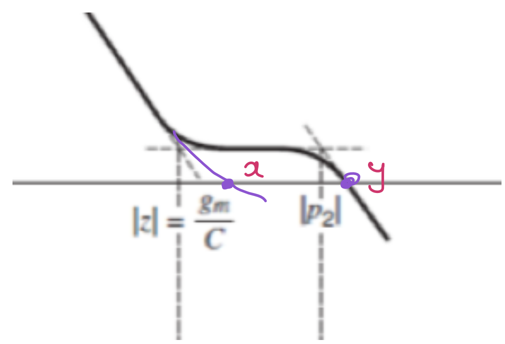
→ point A represents phase loss associated with P_1

point B represents phase loss associated with zero $\boxed{z = \frac{g_m}{C}}$

point C represents phase loss associated with pole P_2

At C The circuit is already out of phase.

→ The effect of the zero is that we are moving the value of ω_{0dB} to higher frequencies.



x: ω_{0dB} without zero

y: ω_{0dB} with zero

∴ Because of the zero there is a unstability issue

→ Situation would be radically different if P_2 were to be less than zero

It's hard to say if P_2 is smaller (or) larger than zero. Typically these are closely placed values.

Q How can we enforce P_2 to be smaller than zero?

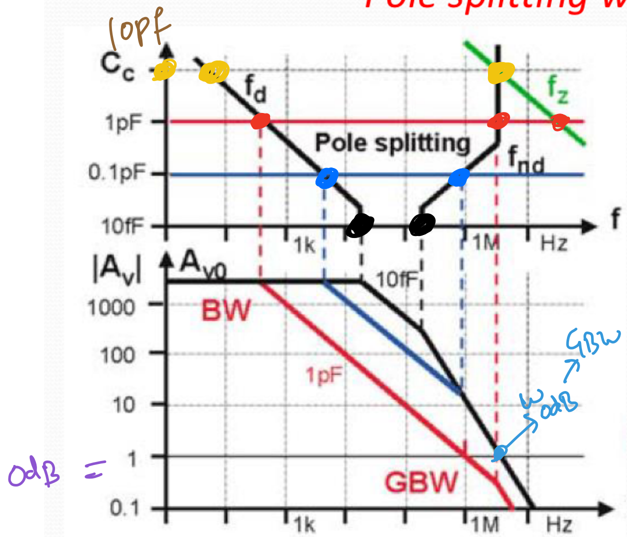
well Basically we can play with circuit parameters. g_m, C_2, C_1

$$p_2 \simeq -\frac{g_m C}{C_2C_1 + C(C_2 + C_1)} \qquad z = \frac{g_{m6}}{C}$$

** A plot that represents Big picture about Poles and zero positioning.

Pole splitting with Cc

Yellow case



$$p_1 \simeq -\frac{1}{g_{m6} R_2 R_1 C}$$

$$z = \frac{g_{m6}}{C}$$

Decrease with increasing C

$$p_2 \simeq -\frac{g_{m6} C}{C_2C_1 + C(C_2 + C_1)}$$

When C is large then

$$p_2 \sim -\frac{g_{m6}}{C_1 + C_2}$$

f_d: Frequency associated with Dominant pole

f_{nd}: Frequency " " 2^{nd} pole

f_z: " " " Zero

C_c: Miller Cos Compensation capacitance.

In the above graphs we are comparing the effect of C_c on pole splitting and in turn on the stability of the circuit.

(1:15:00 – end)

* Black for $C_c = 10fF$

The system looks Unstable

* Blue for $C_c = 0.1\,pF$

The system looks Unstable because we have already crossed two poles and $\varphi_{PM} < 45°$ which is risky.

Zero at High Freq. because of low C_c

* RED for $C_c = 1pF$

but in this case zero is not negligible

The 2^{nd} pole is placed after ω_{0dB}

* Yellow for $C_c = 10pf$

Here the 2^{nd} pole and zero coincide

Q What would happen is zero is at a lower position than 2^{nd} pole?

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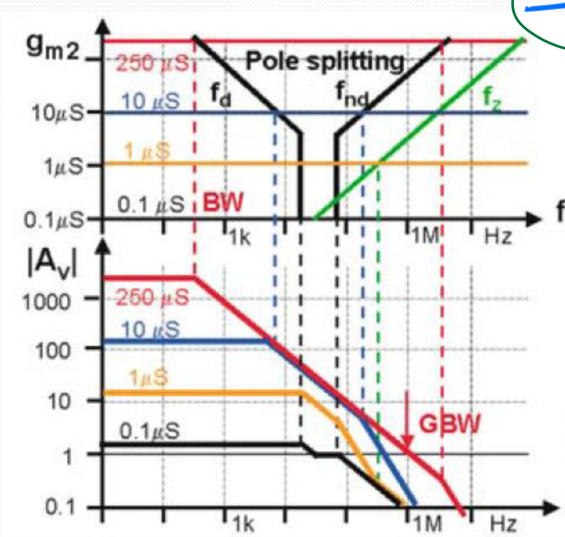
* Additional way to control poles and zeros

Pole splitting with gm 2nd stage

Transconductance of 2^{nd} stage



$$p_1 \simeq -\frac{1}{g_m R_2 R_1 C}$$

$$p_2 \simeq -\frac{g_m C}{C_2 C_1 + C(C_2 + C_1)}$$

$$z = \frac{g_m}{C}$$

RED is most stable.

(GM and W. Sansen)

By

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